Canonical Structural Decomposition and Orthogonalization of Dynamical Modes in Measure Evolution

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Abstract

In this work, I formulate the structural decomposition of evolving densities in its affine form. Instead of separating only mass, translation, and isotropic scale, I separate mass, translation, a global linear deformation, and a normalized residual shape. In these affine coordinates, I define transport and informational projections, prove uniqueness of the structural decomposition under the canonical affine gauge, show that the kinetic action diagonalizes once the residual transport is projected orthogonally to the affine modes, and establish an abstract structural closure principle for the projected residual dynamics. I also explain how the isotropic theory arises as the special case $A(t) = \sigma(t)I$ and record the one-dimensional numerical verification already implemented in the companion Python script as evidence for that special case.

1 Introduction

My goal is to describe density evolution in coordinates that separate global affine motion from genuinely internal deformation. In many formulations, transport, dilation, anisotropic stretching, and residual shape change are mixed together. I want a coordinate system in which these effects are separated as canonically as possible.

The central choice in the present version is that the global geometric part is not merely isotropic. I treat the affine component itself as structural. This leads naturally to a decomposition into:

- mass;
- translation;
- global linear deformation;
- residual shape.

The isotropic formulation remains important, but here I treat it as the special case in which the affine matrix is a scalar multiple of the identity.

2 Affine Structural Variables

Let $\rho(x,t) \geq 0$ be a density on $\mathbb{R}^d$ with finite mass and finite second moment. Fix a symmetric positive definite reference matrix $\Sigma_0 \in \mathbb{R}^{d\times d}$.

Definition 1 (Affine structural variables)**.** *I define the total mass*

$$M(t) = \int_{\mathbb{R}^d} \rho(x,t)\,dx$$

and the barycenter

$$q(t) = \frac{1}{M(t)} \int_{\mathbb{R}^d} x\, \rho(x,t)\, dx.$$

I then define the centered covariance matrix

$$C_\rho(t) = \frac{1}{M(t)} \int_{\mathbb{R}^d} (x - q(t)) \otimes (x - q(t))\, \rho(x,t)\, dx.$$

An affine structural matrix $A(t) \in \mathbb{R}^{d\times d}$ *with* $\det A(t) > 0$ *is any matrix satisfying*

$$A(t)\Sigma_0 A(t)^\top = C_\rho(t).$$

To obtain uniqueness, I impose a gauge choice.

Definition 2 (Canonical affine gauge)**.** *I call the affine structural matrix canonical when* $A(t)$ *is symmetric positive definite. In that case,*

$$A(t) = \Sigma_0^{-1/2}\big(\Sigma_0^{1/2} C_\rho(t) \Sigma_0^{1/2}\big)^{1/2} \Sigma_0^{-1/2}.$$

3 Affine Structural Decomposition

I define the normalized coordinate

$$\xi = A(t)^{-1}(x - q(t))$$

and the residual shape

$$\eta(\xi,t) = \frac{\det A(t)}{M(t)}\, \rho(q(t) + A(t)\xi, t).$$

Proposition 1 (Affine normalization)**.** *The normalized profile satisfies*

$$\int_{\mathbb{R}^d} \eta(\xi,t)\, d\xi = 1, \qquad \int_{\mathbb{R}^d} \xi\, \eta(\xi,t)\, d\xi = 0, \qquad \int_{\mathbb{R}^d} \xi \otimes \xi\, \eta(\xi,t)\, d\xi = \Sigma_0.$$

Proof. The identities follow directly from the substitution $x = q(t) + A(t)\xi$ and the defining relation

$$C_\rho(t) = A(t)\Sigma_0 A(t)^\top.$$

□

4 Functional Setting

To make the construction precise, I fix a functional setting in which the decomposition and the projections are meaningful.

Definition 3 (Admissible densities)**.** *I call* $\rho(\cdot,t)$ *admissible if*

$$\rho(\cdot,t) \geq 0, \qquad \rho(\cdot,t) \in L^1(\mathbb{R}^d), \qquad \int_{\mathbb{R}^d} |x|^2 \rho(x,t)\, dx < \infty,$$

with

$$M(t) > 0 \qquad \text{and} \qquad C_\rho(t) \text{ positive definite.}$$

For time-dependent problems on $[0,T]$*, I work with*

$$\rho \in L^\infty\Big(0,T; L^1(\mathbb{R}^d)\Big) \cap L^\infty\Big(0,T; L^1(\mathbb{R}^d, |x|^2 dx)\Big).$$

Definition 4 (Normalized shape class)**.** *I denote by*

$$\mathcal{S}_{\Sigma_0} = \left\{ \eta \geq 0 : \int \eta \, d\xi = 1, \ \int \xi \eta \, d\xi = 0, \ \int \xi \otimes \xi \, \eta \, d\xi = \Sigma_0 \right\}$$

the class of affine-normalized residual shapes.

Definition 5 (Weighted residual spaces)**.** *Given $\eta \geq 0$, I define*

$$L^2(\eta \, d\xi) = \left\{ f : \int_{\mathbb{R}^d} \eta(\xi) |f(\xi)|^2 \, d\xi < \infty \right\}.$$

For vector fields, I use

$$L^2(\eta \, d\xi; \mathbb{R}^d) = \left\{ w : \int_{\mathbb{R}^d} \eta(\xi) |w(\xi)|^2 \, d\xi < \infty \right\}.$$

5 Existence and Uniqueness of the Affine Decomposition

Theorem 1 (Existence and uniqueness of the affine structural decomposition)**.** *Let $\rho \geq 0$ be an admissible density at a fixed time. Then there exists a unique quadruple*

$$(M, q, A, \eta)$$

such that

$$\rho(x) = \frac{M}{\det A} \, \eta\big(A^{-1}(x - q)\big) \, ,$$

where

$$M > 0, \qquad q \in \mathbb{R}^d, \qquad A \in \mathbb{R}^{d \times d} \text{ is symmetric positive definite}, \qquad \eta \in \mathcal{S}_{\Sigma_0}.$$

Proof. The mass M and barycenter q are uniquely determined by ρ. The centered covariance matrix C_ρ is also uniquely determined. Since C_ρ and Σ_0 are positive definite, the canonical affine matrix

$$A = \Sigma_0^{-1/2} \big(\Sigma_0^{1/2} C_\rho \Sigma_0^{1/2} \big)^{1/2} \Sigma_0^{-1/2}$$

is well defined, symmetric positive definite, and satisfies

$$A \Sigma_0 A^\top = C_\rho.$$

Defining

$$\eta(\xi) = \frac{\det A}{M} \rho(q + A\xi)$$

gives the required normalization identities. Uniqueness follows because M, q, and C_ρ are uniquely determined by ρ, hence so is the canonical matrix A, and then so is η. $\square$

6 Residual Dynamics

In affine coordinates, I write the residual evolution as

$$\partial_t \eta + \nabla_\xi \cdot (\eta w^\perp) = 0,$$

where $w^\perp$ is the projected residual transport field.

For the informational mode, I take as primary definition the weak factorization

$$\partial_t \eta = \eta \psi$$

in the sense of distributions. When $\eta > 0$ almost everywhere and the time derivative is pointwise meaningful, this becomes

$$\psi = \partial_t \log \eta.$$

Proposition 2 (Compatibility of transport and informational residual dynamics)**.** *Assume*

$$\eta \in L^\infty\Big(0, T; L^1(\mathbb{R}^d)\Big), \qquad w^\perp \in L^2\Big(0, T; L^2(\eta\, d\xi; \mathbb{R}^d)\Big),$$

and that

$$\partial_t \eta + \nabla_\xi \cdot (\eta w^\perp) = 0$$

holds in the sense of distributions. If there exists $\psi \in L^2(0, T; L^2(\eta\, d\xi))$ such that

$$\partial_t \eta = \eta \psi$$

in the sense of distributions, then

$$\eta \psi = -\nabla_\xi \cdot (\eta w^\perp)$$

in $\mathcal{D}'(\mathbb{R}^d \times (0, T))$. Conversely, if

$$-\nabla_\xi \cdot (\eta w^\perp) = \eta \psi$$

for some $\psi \in L^2(0, T; L^2(\eta\, d\xi))$, then the informational evolution holds automatically.

Proof. Both equations identify the same distribution $\partial_t \eta$. Subtracting them gives

$$\eta \psi + \nabla_\xi \cdot (\eta w^\perp) = 0$$

in the sense of distributions. The converse is immediate by substitution. □

7 Affine Projection of Transport Modes

The affine modes are exactly the constant and linear vector fields in ξ.

Definition 6 (Projected affine transport mode)**.** *Let $w \in L^2(\eta\, d\xi; \mathbb{R}^d)$. I define*

$$w^\perp(\xi, t) = w(\xi, t) - a(t) - B(t)\xi,$$

where

$$a(t) = \int_{\mathbb{R}^d} \eta(\xi, t) w(\xi, t)\, d\xi$$

and

$$B(t) = \left(\int_{\mathbb{R}^d} \eta(\xi, t)\, w(\xi, t) \otimes \xi\, d\xi \right) \Sigma_0^{-1}.$$

Proposition 3 (Affine transport orthogonality)**.** *The projected transport field satisfies*

$$\int_{\mathbb{R}^d} \eta(\xi,t) w^{\perp}(\xi,t)\, d\xi = 0$$

and

$$\int_{\mathbb{R}^d} \eta(\xi,t)\, \xi \otimes w^{\perp}(\xi,t)\, d\xi = 0.$$

Proof. These identities follow immediately from the definitions of $a(t)$ and $B(t)$ and from the normalization

$$\int \xi \otimes \xi\, \eta\, d\xi = \Sigma_0.$$

□

8 Projection of Informational Modes

Definition 7 (Projected informational mode)**.** *I define*

$$\psi^{\perp}(\xi,t) = \psi(\xi,t) - \alpha(t) - \beta(t)\cdot\xi - (\xi\otimes\xi - \Sigma_0) : \Gamma(t),$$

where $\alpha(t) \in \mathbb{R}$, $\beta(t) \in \mathbb{R}^d$, *and* $\Gamma(t) \in \mathbb{R}^{d\times d}_{\mathrm{sym}}$ *are chosen so that*

$$\int \eta\psi^{\perp}\, d\xi = 0, \qquad \int \xi\, \eta\psi^{\perp}\, d\xi = 0, \qquad \int (\xi\otimes\xi - \Sigma_0)\, \eta\psi^{\perp}\, d\xi = 0.$$

I interpret the last condition componentwise as the vanishing of a symmetric matrix of quadratic moments.

This is the scalar affine counterpart of the transport projection, written in the centered quadratic basis naturally associated with the affine normalization.

9 Velocity Field in Physical Coordinates

Let

$$u(t) = \log M(t).$$

The affine macro velocity generated by $q(t)$ and $A(t)$ is

$$v_{\mathrm{macro}}(x,t) = \dot{q}(t) + \dot{A}(t)A(t)^{-1}(x - q(t)).$$

The full physical velocity field is then

$$v(x,t) = \dot{q}(t) + \dot{A}(t)A(t)^{-1}(x - q(t)) + A(t)w^{\perp}(\xi,t).$$

10 Diagonalization of the Kinetic Action

Theorem 2 (Affine diagonalization of the kinetic action)**.** *Under the affine normalization and projection conditions,*

$$\int_{\mathbb{R}^d} \rho(x,t)|v(x,t)|^2\, dx = M(t)|\dot{q}(t)|^2 + M(t)\,\mathrm{tr}\Big(\dot{A}(t)\Sigma_0\dot{A}(t)^{\top}\Big) + M(t)\int_{\mathbb{R}^d} \eta(\xi,t)\,|A(t)w^{\perp}(\xi,t)|^2\, d\xi.$$

In particular, all mixed terms between translation, affine deformation, and residual transport vanish.

Proof. Using $x - q = A\xi$ and $\rho(x,t)\,dx = M\eta(\xi,t)\,d\xi$, I rewrite the velocity as

$$v = \dot{q} + \dot{A}\,\xi + Aw^{\perp}.$$

Expanding $|v|^2$ produces the mixed terms

$$\dot{q}\cdot\dot{A}\,\xi, \qquad \dot{q}\cdot Aw^{\perp}, \qquad \dot{A}\,\xi\cdot Aw^{\perp}.$$

After integration, these are proportional to

$$\int \xi\,\eta\,d\xi, \qquad \int \eta w^{\perp}\,d\xi, \qquad \int \eta\,\xi\otimes w^{\perp}\,d\xi,$$

all of which vanish by normalization and affine projection.

The diagonal affine term is

$$M\int \eta\,|\dot{A}\,\xi|^2\,d\xi = M\,\mathrm{tr}\Big(\dot{A}\Sigma_0\dot{A}^{\top}\Big).$$

This yields the formula. □

11 Action Functional

I define the total action by

$$\mathcal{A}_{\mathrm{tot}} = \mathcal{A}_{\mathrm{macro}} + \mathcal{A}_{\mathrm{micro,tr}} + \mathcal{A}_{\mathrm{micro,info}}.$$

11.1 Affine macroscopic component

$$\mathcal{A}_{\mathrm{macro}} = \int_0^1 \Big[\lambda_M \dot{u}(t)^2 + \lambda_q|\dot{q}(t)|^2 + \lambda_A\,\mathrm{tr}\Big(\dot{A}(t)\Sigma_0\dot{A}(t)^{\top}\Big)\Big]\,dt.$$

11.2 Transport residual

$$\mathcal{A}_{\mathrm{micro,tr}} = \nu_{\mathrm{tr}}\int_0^1\int_{\mathbb{R}^d} \eta(\xi,t)|A(t)w^{\perp}(\xi,t)|^2\,d\xi\,dt.$$

11.3 Informational residual

$$\mathcal{A}_{\mathrm{micro,info}} = \nu_{\mathrm{info}}\int_0^1\int_{\mathbb{R}^d} \eta(\xi,t)|\psi^{\perp}(\xi,t)|^2\,d\xi\,dt.$$

12 Affine Projection Theory

Proposition 4 (Linear independence of the affine informational basis)**.** *Assume $\eta \geq 0$, $\eta \not\equiv 0$, and that the support of η is not contained in any proper affine quadratic zero set. Then the family*

$$\{1, \xi_1, \ldots, \xi_d, \xi_i\xi_j - (\Sigma_0)_{ij} : 1 \leq i \leq j \leq d\}$$

is linearly independent in $L^2(\eta\,d\xi)$.

Proof. If a quadratic polynomial vanishes in $L^2(\eta\,d\xi)$, then it vanishes η-almost everywhere, so the support of η lies inside its zero set. By assumption, this is possible only for the zero polynomial. □

Proposition 5 (Positive definiteness of the affine informational Gram matrix)**.** *If $\eta \geq 0$, $\int \eta \, d\xi > 0$, the moments up to order four are finite, and the affine informational basis above is linearly independent in $L^2(\eta \, d\xi)$, then the corresponding weighted Gram matrix is symmetric positive definite.*

Proof. For any quadratic polynomial p in the affine informational basis,

$$\int \eta |p|^2 \, d\xi \geq 0,$$

with equality only when $p = 0$ in $L^2(\eta \, d\xi)$. By linear independence, this implies positive definiteness. □

Proposition 6 (Uniqueness of the weighted affine transport projection)**.** *Let $\eta \in \mathcal{S}_{\Sigma_0}$ with Σ_0 symmetric positive definite, and let*

$$w \in L^2(\eta \, d\xi; \mathbb{R}^d).$$

Then the coefficients $a \in \mathbb{R}^d$ and $B \in \mathbb{R}^{d \times d}$ defining

$$w^{\perp}(\xi) = w(\xi) - a - B\xi$$

through the orthogonality conditions

$$\int \eta w^{\perp} \, d\xi = 0, \qquad \int \eta \, \xi \otimes w^{\perp} \, d\xi = 0$$

are uniquely determined.

Proof. The first condition forces

$$a = \int \eta w \, d\xi$$

because $\int \xi \, \eta \, d\xi = 0$. The second condition becomes

$$\left(\int \eta \, \xi \otimes w \, d\xi \right) - \Sigma_0 B^{\top} = 0.$$

Since Σ_0 is symmetric positive definite, it is invertible, and therefore

$$B^{\top} = \Sigma_0^{-1} \left(\int \eta \, \xi \otimes w \, d\xi \right)$$

is uniquely determined. □

Proposition 7 (Uniqueness of the weighted affine informational projection)**.** *Under the assumptions of the affine informational Gram proposition, the coefficients of*

$$\psi^{\perp}(\xi) = \psi(\xi) - \alpha - \beta \cdot \xi - (\xi \otimes \xi - \Sigma_0) : \Gamma$$

are uniquely determined by the defining orthogonality conditions.

Proof. The coefficients are obtained by solving a finite-dimensional linear system with the affine informational Gram matrix. Positive definiteness gives invertibility and uniqueness. □

13 Stability Under Convergent Density Sequences

Proposition 8 (Tightness from mass and second-moment bounds)**.** *Let $(\rho_n)_n$ be a sequence of nonnegative densities such that*

$$\sup_n \int \rho_n \, dx < \infty, \qquad \sup_n \int |x|^2 \rho_n(x) \, dx < \infty.$$

Then $(\rho_n)_n$ is tight.

Proof. For every $R > 0$,

$$\int_{|x|>R} \rho_n(x) \, dx \le \frac{1}{R^2} \int |x|^2 \rho_n(x) \, dx.$$

Taking the supremum over n proves the claim. □

Theorem 3 (Stability of the affine normalization under L^1 convergence)**.** *Let $\rho_n \to \rho$ in $L^1(\mathbb{R}^d)$ and assume*

$$\sup_n \int |x|^2 \rho_n(x) \, dx < \infty.$$

Let (M_n, q_n, A_n, η_n) and (M, q, A, η) be the associated canonical affine decompositions, and assume

$$M > 0, \qquad C_\rho \text{ positive definite}, \qquad C_{\rho_n} \to C_\rho.$$

Then

$$M_n \to M, \qquad q_n \to q, \qquad A_n \to A, \qquad \eta_n \to \eta \quad \text{in } L^1(\mathbb{R}^d).$$

Proof. The mass convergence follows from L^1 convergence. Tightness gives convergence of first moments, hence $q_n \to q$. The convergence $C_{\rho_n} \to C_\rho$ and the continuity of the matrix square root on positive definite matrices imply

$$A_n = \Sigma_0^{-1/2} \big(\Sigma_0^{1/2} C_{\rho_n} \Sigma_0^{1/2}\big)^{1/2} \Sigma_0^{-1/2} \to \Sigma_0^{-1/2} \big(\Sigma_0^{1/2} C_\rho \Sigma_0^{1/2}\big)^{1/2} \Sigma_0^{-1/2} = A.$$

Writing

$$\eta_n(\xi) = \frac{\det A_n}{M_n} \rho_n(q_n + A_n \xi), \qquad \eta(\xi) = \frac{\det A}{M} \rho(q + A\xi),$$

I split the difference

$$\eta_n - \eta$$

into a term controlled directly by $\|\rho_n - \rho\|_{L^1}$ after the change of variables $x = q_n + A_n \xi$, and a term controlled by continuity of translations and invertible linear changes of variables in L^1. This yields $\eta_n \to \eta$ in L^1. □

Proposition 9 (Convergence of affine projection coefficients)**.** *Assume the hypotheses of the previous theorem. Let*

$$w_n \in L^2(\eta_n \, d\xi; \mathbb{R}^d), \qquad w \in L^2(\eta \, d\xi; \mathbb{R}^d),$$

and let (a_n, B_n) and (a, B) be the corresponding affine transport projection coefficients. Assume

$$\int \eta_n w_n \, d\xi \to \int \eta w \, d\xi, \qquad \int \eta_n \, \xi \otimes w_n \, d\xi \to \int \eta \, \xi \otimes w \, d\xi.$$

Then

$$a_n \to a, \qquad B_n \to B.$$

If, in addition, $\psi_n \in L^2(\eta_n\, d\xi)$ and $\psi \in L^2(\eta\, d\xi)$,

$$\sup_n \int_{\mathbb{R}^d} |\xi|^4 \eta_n(\xi)\, d\xi < \infty$$

and the affine informational moment vectors built from

$$\int \eta_n \psi_n\, d\xi, \qquad \int \xi\, \eta_n \psi_n\, d\xi, \qquad \int (\xi \otimes \xi - \Sigma_0)\, \eta_n \psi_n\, d\xi$$

converge to their limits for (η, ψ), and the affine informational Gram matrices are uniformly coercive, then the informational projection coefficients converge as well.

Proof. The transport coefficients are explicit affine moments of the pairs (η_n, w_n), so the assumed convergence of those moments gives $a_n \to a$ and $B_n \to B$. The informational coefficients solve finite-dimensional linear systems

$$G_n c_n = b_n.$$

The fourth-moment bound guarantees that all entries of G_n and b_n are well defined and uniformly controlled. Uniform coercivity gives bounded inverses, and the convergence of the corresponding moments gives $G_n \to G$ and $b_n \to b$. □

Proposition 10 (Quantitative stability of the affine structural map)**.** *Let $\rho_1, \rho_2 \geq 0$ be admissible densities with canonical affine decompositions*

$$\rho_i(x) = \frac{M_i}{\det A_i}\, \eta_i\big(A_i^{-1}(x - q_i)\big), \qquad i = 1, 2.$$

Assume that

$$0 < m_* \leq M_i \leq m^*, \qquad \int_{\mathbb{R}^d} |x|^2 \rho_i(x)\, dx \leq K_2,$$

and that their centered covariance matrices satisfy

$$\lambda I \leq C_{\rho_i} \leq \Lambda I.$$

Assume also that

$$\|\rho_1 - \rho_2\|_{L^1(\mathbb{R}^d)} < \infty, \qquad \int_{\mathbb{R}^d} |x|\, |\rho_1 - \rho_2|(x)\, dx < \infty.$$

Then there exists a constant

$$C = C(m_*, m^*, K_2, \lambda, \Lambda, \Sigma_0, d)$$

such that

$$|M_1 - M_2| + |q_1 - q_2| + \|A_1 - A_2\| \leq C \left(\|\rho_1 - \rho_2\|_{L^1} + \int |x|\, |\rho_1 - \rho_2|(x)\, dx + \|C_{\rho_1} - C_{\rho_2}\| \right).$$

Moreover, on every uniformly admissible class satisfying the same bounds, the map

$$\rho \longmapsto (M, q, A, \eta)$$

is locally Lipschitz from the metric induced by

$$\|\rho\|_{L^1} + \int |x|\, \rho(x)\, dx + \|C_\rho\|$$

into

$$\mathbb{R}_+ \times \mathbb{R}^d \times \mathbb{R}^{d \times d} \times L^1(\mathbb{R}^d).$$

Proof. The mass estimate is immediate from

$$|M_1 - M_2| = \left| \int (\rho_1 - \rho_2)\, dx \right| \le \|\rho_1 - \rho_2\|_{L^1}.$$

For the barycenters, I write

$$q_i = \frac{1}{M_i} \int x\, \rho_i(x)\, dx$$

and separate the contributions of the masses and of the first moments. This gives a bound by

$$C \left(\|\rho_1 - \rho_2\|_{L^1} + \int |x|\, |\rho_1 - \rho_2|(x)\, dx \right).$$

The centered covariance matrices vary continuously under the same controls because the second moments are uniformly bounded. The canonical affine matrices are given by

$$A_i = \Sigma_0^{-1/2} \big(\Sigma_0^{1/2} C_{\rho_i} \Sigma_0^{1/2} \big)^{1/2} \Sigma_0^{-1/2},$$

and the matrix square root is locally Lipschitz on uniformly positive definite matrices. Hence

$$\|A_1 - A_2\| \le C \|C_{\rho_1} - C_{\rho_2}\|.$$

Finally, the residual profiles are written as

$$\eta_i(\xi) = \frac{\det A_i}{M_i} \rho_i(q_i + A_i \xi),$$

so they vary continuously under the same controls by continuity of translations and invertible linear changes of variables in L^1 on uniformly admissible classes. This proves the local Lipschitz claim. □

14 Abstract Admissible Dynamics

Theorem 4 (Abstract consistency of an admissible affine projected dynamics)**.** *Let*

$$\eta_0 \in \mathcal{S}_{\Sigma_0} \cap L^1(\mathbb{R}^d, (1 + |\xi|^4) d\xi) \cap L^2(\mathbb{R}^d).$$

Assume there exist

$$\eta \in L^\infty \Big(0, T; L^1(\mathbb{R}^d, (1 + |\xi|^4) d\xi) \cap L^2(\mathbb{R}^d) \Big),$$

$$w \in L^2 \Big(0, T; L^2(\eta\, d\xi; \mathbb{R}^d) \Big), \qquad \psi \in L^2 \big(0, T; L^2(\eta\, d\xi) \big),$$

such that

$$\partial_t \eta + \nabla_\xi \cdot (\eta w^\perp) = 0, \qquad \partial_t \eta = \eta \psi$$

hold in the sense of distributions with $\eta(\cdot, 0) = \eta_0$, where $w^\perp$ is the affine projection of w. Assume also:

1. *the affine informational Gram matrices are positive definite for almost every t;*
2. *the fluxes $\eta w^\perp$, $|\xi| \eta w^\perp$, and $|\xi|^2 \eta w^\perp$ are integrable and decay sufficiently at infinity;*

Then the affine projected fields $w^\perp$ and $\psi^\perp$ are well defined for almost every t and satisfy

$$w^\perp \in L^2\Big(0,T;L^2(\eta\,d\xi;\mathbb{R}^d)\Big)\,, \qquad \psi^\perp \in L^2\big(0,T;L^2(\eta\,d\xi)\big)\,.$$

Moreover, the affine normalization constraints remain preserved for all $t \in [0,T]$.

Proof. The affine covariance normalization gives a well-defined transport projection, and the affine informational Gram positivity gives a well-defined informational projection. The fourth-moment bound on η ensures that the quadratic correction defining $\psi^\perp$ still belongs to $L^2(\eta\,d\xi)$. Since the evolution is driven by $w^\perp$, the affine transport orthogonality holds by construction at almost every time. Testing the continuity equation against 1, ξ, and $\xi\otimes\xi$ gives preservation of mass, barycenter, and affine covariance normalization. □

Proposition 11 (Abstract structural closure principle for the affine residual dynamics)**.** *Suppose a concrete evolution problem is given in a class of admissible paths*

$$\eta \in L^\infty\Big(0,T;L^1(\mathbb{R}^d,(1+|\xi|^4)d\xi)\cap L^2(\mathbb{R}^d)\Big)$$

for which:

1. *for every admissible initial datum*

$$\eta_0 \in \mathcal{S}_{\Sigma_0}\cap L^1(\mathbb{R}^d,(1+|\xi|^4)d\xi)\cap L^2(\mathbb{R}^d)$$

   there exists a distributional solution η;

2. *the induced transport field w belongs to $L^2(0,T;L^2(\eta\,d\xi;\mathbb{R}^d))$;*

3. *the induced source satisfies $-\nabla_\xi\cdot(\eta w^\perp)=\eta\psi$ for some $\psi\in L^2(0,T;L^2(\eta\,d\xi))$;*

4. *the affine informational Gram matrices remain positive definite almost everywhere in time.*

Then the affine projected dynamics is structurally closed in that class: the projected fields are uniquely determined almost everywhere, the normalization constraints are propagated, and the decomposition remains admissible for all times for which the underlying evolution exists.

Proof. Existence of the path is provided by the underlying evolution problem. Proposition 2 identifies the transport and informational descriptions of the same residual evolution. The projection uniqueness results give unique coefficients almost everywhere in time, and Theorem 4 propagates the affine normalization constraints. □

Theorem 5 (Propagation of the affine structural decomposition)**.** *Let $\rho(\cdot,t)\geq 0$ be a time-dependent density on $[0,T]$ such that*

$$\rho \in L^\infty\Big(0,T;L^1(\mathbb{R}^d)\cap L^1(\mathbb{R}^d,|x|^2dx)\Big)\,,$$

the mass $M(t)$ is strictly positive for all $t\in[0,T]$, and the centered covariance matrix $C_\rho(t)$ remains positive definite for all $t\in[0,T]$. Assume moreover that the moments of order up to two depend continuously on time. If $\rho(\cdot,0)$ is admissible, then for every $t\in[0,T]$:

1. *the canonical affine decomposition*

$$\rho(x,t) = \frac{M(t)}{\det A(t)}\,\eta\big(A(t)^{-1}(x-q(t)),t\big)$$

is well defined;

2. *the canonical affine gauge remains valid, that is, $A(t)$ is the unique symmetric positive definite matrix satisfying*

$$A(t)\Sigma_0 A(t)^\top = C_\rho(t);$$

3. *the residual profile remains in the normalized class*

$$\eta(\cdot,t) \in \mathcal{S}_{\Sigma_0}.$$

Proof. For each time t, the assumptions guarantee that the mass, barycenter, and centered covariance matrix are well defined, with

$$M(t) > 0 \qquad \text{and} \qquad C_\rho(t) \text{ positive definite.}$$

Therefore the canonical affine matrix

$$A(t) = \Sigma_0^{-1/2}\big(\Sigma_0^{1/2} C_\rho(t)\Sigma_0^{1/2}\big)^{1/2}\Sigma_0^{-1/2}$$

is well defined and unique in the symmetric positive definite class. Defining

$$\eta(\xi,t) = \frac{\det A(t)}{M(t)}\,\rho(q(t)+A(t)\xi,t),$$

the same change of variables used in the static normalization proposition gives

$$\int \eta(\xi,t)\,d\xi = 1, \qquad \int \xi\,\eta(\xi,t)\,d\xi = 0, \qquad \int \xi\otimes\xi\,\eta(\xi,t)\,d\xi = \Sigma_0.$$

Hence the affine decomposition remains well defined, the gauge remains canonical, and the normalization constraints propagate for all times. □

15 Invariant Analysis

Proposition 12 (Affine structural invariants)**.** *Assume*

$$\rho(x,t) = \frac{M(t)}{\det A(t)}\eta\big(A(t)^{-1}(x-q(t)),t\big)\,,$$

with $M(t) > 0$, $\det A(t) > 0$, and $\eta \in \mathcal{S}_{\Sigma_0}$. Then:

- *positivity of η implies positivity of ρ;*
- *the mass of ρ is $M(t)$;*
- *the barycenter of ρ is $q(t)$;*
- *the covariance of ρ is $A(t)\Sigma_0 A(t)^\top$.*

Proof. All four statements follow directly from the affine change of variables $x = q + A\xi$. □

16 A Structural Hierarchy Beyond the Affine Level

The affine CSD should not be read only as an isolated reformulation. The numerical evidence and the failure-mode experiments suggest a broader structural hierarchy in which increasingly rich macrogeometric classes capture increasingly rich classes of deformations. In that hierarchy, the affine CSD occupies the first genuinely nontrivial geometric level: it goes strictly beyond isotropic scaling, but it still remains substantially simpler than quadratic or multimodal descriptions.

I therefore propose the following hierarchy of structural models.

- V_1: isotropic scaling, with macrovariables (M, q, σ) and residual shape transported only up to a scalar scale;
- V_2: affine CSD, with macrovariables (M, q, A) and residual shape transported up to a full linear deformation;
- V_3: a quadratic extension, in which the global transport ansatz is enriched by quadratic corrections in the structural coordinate;
- V_4: a multimodal extension, in which the density is represented by a finite collection of structural components rather than a single residual profile.

At the formal level, each structural level defines a class of admissible reconstructions. Denoting by $\mathcal{C}_1, \mathcal{C}_2, \mathcal{C}_3, \mathcal{C}_4$ the corresponding model classes, I expect the nesting

$$\mathcal{C}_1 \subset \mathcal{C}_2 \subset \mathcal{C}_3 \subset \mathcal{C}_4.$$

If $\mathcal{E}_k(\rho)$ denotes the optimal reconstruction error of ρ within $\mathcal{C}_k$, then the hierarchy predicts the monotonicity

$$\mathcal{E}_4(\rho) \leq \mathcal{E}_3(\rho) \leq \mathcal{E}_2(\rho) \leq \mathcal{E}_1(\rho).$$

The hierarchy is useful only if it also explains when I should move from one level to the next. The guiding principle suggested by the experiments is the following. I should remain at level V_2 when the affine reconstruction error is already significantly smaller than the isotropic one and the residual shape remains essentially unimodal and weakly curved. I should move from V_2 to V_3 when the affine level improves over V_1 but leaves a persistent curved or folded residual pattern. I should move from V_2 or V_3 to V_4 when the residual shape becomes genuinely fragmented or multimodal and no single global component remains sufficient.

The role of the present manuscript is to develop the level V_2 in a canonical way. The isotropic theory appears as V_1, recovered when $A = \sigma I$. The experiments developed outside the manuscript indicate why levels beyond V_2 are natural. In strongly anisotropic but essentially linear regimes, the affine level already captures most of the improvement over isotropic scaling. In smoothly folded regimes, quadratic corrections improve substantially over the affine level. In fragmented or genuinely multimodal regimes, multimodal extensions improve further. By contrast, in nearly isotropic regimes the gain of V_2 over V_1 may be modest or may disappear.

This hierarchy gives a principled interpretation of the affine CSD. It is not merely a special-purpose model between two unrelated descriptions. Rather, it is the first structural level at which global deformation is treated geometrically instead of being hidden inside the residual. For that reason, I regard the affine CSD as the central level of the structural hierarchy: it is the minimal level that separates translation, global deformation, and internal deformation in a canonical way, while still remaining analytically tractable.

16.1 Adaptive Structural Hierarchy

I regard the affine CSD developed in this manuscript as the official level V_2 of a broader adaptive structural hierarchy. In that hierarchy, the isotropic theory is the level V_1, while the next natural extensions are:

- V_3, a quadratic structural extension in which the global transport ansatz is enriched by quadratic corrections in the structural coordinate;
- V_4, a multimodal structural extension in which the density is represented by a finite collection of structural components instead of a single residual profile.

The purpose of this hierarchy is not to replace the affine CSD, but to place it at the central decision level. The affine level should be used whenever a single global linear deformation already captures the dominant geometry of the evolving density. One should move from V_2 to V_3 when the affine approximation improves substantially over V_1 but leaves a residual that is still essentially unimodal and exhibits persistent curvature or folding. One should move from V_2 or V_3 to V_4 when the residual becomes genuinely fragmented or multimodal and no single global component remains sufficient.

At the formal level, this suggests an adaptive model-selection principle based on the structural errors

$$\mathcal{E}_1(\rho), \qquad \mathcal{E}_2(\rho), \qquad \mathcal{E}_3(\rho), \qquad \mathcal{E}_4(\rho),$$

where $\mathcal{E}_k(\rho)$ denotes the optimal reconstruction error of ρ within the k-th structural class. The hierarchy should then be traversed incrementally: first compare V_1 with V_2; if the gain of V_2 is insufficient or if the residual geometry remains strongly curved, compare V_2 with V_3; if fragmentation still dominates, compare V_3 with V_4. In this way, the affine CSD becomes the central adaptive layer of the theory rather than merely one approximation among many.

17 Isotropic Case as a Special Affine Regime

The isotropic formulation appears when

$$A(t) = \sigma(t)I.$$

If I choose

$$\Sigma_0 = \mu_2 I,$$

then the affine decomposition reduces to

$$\rho(x,t) = \frac{M(t)}{\sigma(t)^d}\eta\left(\frac{x-q(t)}{\sigma(t)}, t\right),$$

and the affine macro velocity reduces to

$$\dot{q}(t) + \dot{\sigma}(t)\sigma(t)^{-1}(x-q(t)) = \dot{q}(t) + \dot{s}(t)(x-q(t)), \qquad s(t) = \log\sigma(t).$$

Thus the earlier isotropic theory is recovered exactly as the special case in which the global linear deformation is scalar.

17.1 One-dimensional numerical verification

The companion script `verify_conjectures.py` implements the isotropic one-dimensional case, which in one dimension coincides with the full affine theory. For that implementation, executed with Python 3.14.3, NumPy 2.4.3, and SciPy 1.17.1, I obtained:

$$\max_{t\in[0,1]} E_{\mathrm{diag}}(t) = 1.221245 \times 10^{-15}, \qquad \max_{t\in[0,1]} E_{\mathrm{cross}}(t) = 1.353084 \times 10^{-16},$$

with transport orthogonality error

$$1.769418 \times 10^{-16},$$

informational orthogonality error

$$1.947227 \times 10^{-16},$$

and final state

$$M(1) = 1.051271, \qquad q(1) = 0.050000, \qquad \sigma(1) = 1.191616.$$

In that script, the informational projection is written in the equivalent uncentered basis $\{1, \xi, \xi^2\}$; in one dimension this spans the same quadratic space as the centered affine basis $\{1, \xi, \xi^2 - \mu_2\}$. This does not prove the full affine theory numerically in arbitrary dimension, but it does verify the one-dimensional special case at machine precision.

18 Conclusion

In this work, I formulated the structural decomposition of evolving densities in affine form. The main point is that the global linear deformation belongs to the macrostructure of the density and should not be hidden inside the residual. With that choice, the natural residual transport is the orthogonal complement of the affine modes, and the kinetic action diagonalizes accordingly.

The present version supports the following main claims:

- the affine structural decomposition is well defined and unique under the canonical gauge;
- the transport projection is uniquely determined by the affine covariance normalization, and the informational projection is uniquely determined under nondegeneracy of the affine informational Gram matrix;
- the kinetic action diagonalizes in affine structural coordinates;
- the transport and informational residual formulations are compatible whenever both are defined in the adopted weak sense;
- the affine residual dynamics is structurally closed at the abstract level once an underlying admissible evolution is given;
- the isotropic formulation is recovered as the special case $A = \sigma I$;
- the companion numerical implementation confirms the one-dimensional special case at machine precision.

I regard multidimensional numerical experiments, concrete PDE existence theorems, and external empirical datasets as natural next steps, but I do not claim them as completed results in the present manuscript.

Declarations

Funding I received no financial support for this research.

Conflicts of interest I declare that there is no conflict of interest.

Data availability No external datasets were analyzed in the present version of this study.

Code availability The numerical verification script `verify_conjectures.py` is provided with the manuscript source and the associated code is made available under the GNU AGPLv3 license in the project repository. The reported numerical experiment was executed with Python 3.14.3, NumPy 2.4.3, and SciPy 1.17.1.

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